Reducing Bank-Synchronization Cycles of a Zero-Skipping FPGA CNN Core by Offline Channel Permutation

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Abstract—A zero-skipping convolution core with lockstep input banks waits, in every window, for the bank with the most nonzero activations. In our integer core on a Zynq XC7Z020, where the channel count is a multiple of 4, the channel order decides which channels share a bank. We fit this order offline and fold it into the adjacent layers, leaving the bitstream and integer output unchanged; the layer shapes show where this cannot pay. At four values per cycle, two relabelings of a 32-channel denoiser cut the core cycles measured on one image (transcribed from run notes) by 7.02% and 7.60%, which a model pushed to GitHub before the export predicted to the cycle net of reported DMA stalls; under a protocol pushed before the fits, the pooled count fell on all ten networks. On this denoiser, a data-free skewed tap layout, counted after the study and not run on the board, saves more: 9.20% pooled over 79 images against 7.46% and 8.10% for the two relabelings.

Keywords—zero skipping, activation sparsity, load balancing, channel permutation, FPGA, CNN accelerator, cycle model

I. INTRODUCTION

A core that skips zero activations saves multiplications, but lanes that advance together make each window cost as much as the busiest one, a stall Cnvlutin measures as its own category [1]. Known remedies for activation imbalance change the network or add hardware: S2TA bounds the nonzeros per block by discarding activations [2], and Sense rebalances channels at run time, noting that input maps "cannot be trained offline to balance IFM workload" [3]. Offline reorderings target weights: SparTen sorts filters by density to balance them and reorders the next layer's weights to match [4], and Pool and Yu fold into the preceding layer a channel permutation that lets N:M pruning keep more weight magnitude [5]. VARA permutes inputs offline from activation statistics to cluster co-zero inputs, reading them through an index unit [6], and skewed storage gives conflict-free parallel access to the rows and columns of a matrix [7]. Permutation, folding, profiling and skewing are thus not new; our search (Crossref, arXiv, web; October 2026) found no work that spreads co-active channels over lockstep banks by an exact, folded relabeling.

In our core, four tap banks run in lockstep, and in 32- and 64-channel layers all nine taps of a channel share the bank given by its index mod 4. A folded channel relabeling moves nonzeros between banks with the function exact, but balances them only on average, as activations change with the input. We give a shape-only condition necessary for a layout to change a layer's cycles (Section II), a board measurement predicted to the cycle (Section V-A), a ten-network study under a protocol pushed to GitHub before the fits, with baselines (Sections V-B, V-C), and a data-free skew of the taps, counted on the board network after the study (Section V-C). The contribution is empirical: how much of the bank-imbalance cost static layouts remove on unseen images, and cycle-exact prediction.

II. CORE, COST MODEL AND BANK LAYOUT

Core. The host builds each layer's windows (im2col), sends them with the layer's weights and requantizes between layers; the core computes every convolution with INT8 weights, 7-bit unsigned activations and INT32 accumulation. Its 64 multipliers form P = 16 output lanes times T = 4 tap banks; its window port takes W values per cycle (W = 2 only in RTL simulation). In sparse mode, used throughout, each bank processes only its nonzero taps.

Cost of a window. For a window with K taps and COUT outputs, and m the largest number of nonzero taps in one bank, the core spends C cycles:

$$C = \max(\lceil K/W \rceil + 1, \lceil COUT/P \rceil \cdot \max(1, m), COUT), \quad (1)$$

for loading the next window, computing $\lceil COUT/P \rceil$ output groups and emitting COUT results. Each layer adds a fill and drain of min(load, compute of the last window) + 3 + g cycles, g being the outputs in the last group of P, a rule read off board runs of four networks; the first layer has a measured constant (17 cycles for the board network at W = 4). We call (1) summed over the windows the count, and the count plus the fill and drain the exact prediction.

Bank of a tap. In (ky, kx, channel) order, tap t = s·N + p (s the 3×3 position, p the channel, N the input channels) sits in bank t mod T, which is p mod 4 when 4 divides N: in this Original (conventional) order each bank holds N/4 whole channels (Fig. 1a), and a permutation of the channels decides which share a bank (Fig. 1b). As the host builds the windows, the taps can also be reordered together with the consumer's weight columns; no partial sum, in any order, exceeds $|bias| + K \cdot 127 \cdot 128 < 2^{31}$, so the INT32 result is unchanged. The data-free skew, skewed storage [7] applied to the taps, puts channel p of position s in slot s·N + (p + s) mod N, bank (p + s) mod 4, spreading each channel's nine taps 3/2/2/2 over the banks (Fig. 1c).

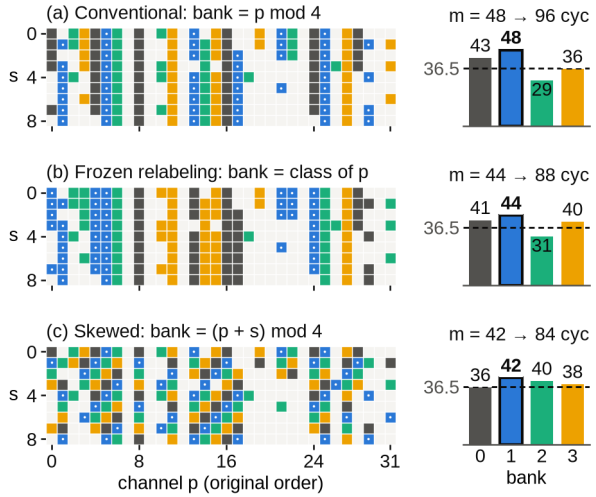


Fig. 1. A conv5 input window of the board network on Set12 05.png (of the windows compute-bound at W = 4, the first in raster order with the median m of 48). Rows: positions s; columns: channels p; filled: nonzero, colored by bank (bank 1 dotted). (a) Original, bank p mod 4; (b) Frozen (16 of 32 channels change bank); (c) skew, bank (p + s) mod 4. Bars: nonzeros per bank; the fullest, m, costs 2m cycles; dashed: m = 36.5, where 2m equals the load of 73 cycles.

When a layout can pay. A bank holds at most $\lceil K/T \rceil$ taps. No window is then compute-bound for any input, and no layout can change the count, if

$$\lceil COUT/P \rceil \cdot \lceil K/T \rceil \le \max(\lceil K/W \rceil + 1, COUT). \quad (2)$$

Failing (2) is thus necessary for a gain; without the rounding, (2) reads COUT ≤ P·T/W, the ridge point of a roofline [8] whose bandwidth roof is the load. So 32-output hidden layers (K = 288: 144 against 145 at W = 2 and 73 at W = 4) can pay only at W = 4, layers with 45 or 64 outputs also at W = 2, and none at W = 1 or in the last layer (COUT ≤ 4). As a reference, perfect balance, m = $\lceil n/T \rceil$ for n nonzero taps, bounds every assignment of a window's taps to lockstep banks.

III. Function-Preserving Relabeling

Folding. Let π map a position p of a hidden map to its original channel. The producer computes its output channels in the order π, with weight rows, biases and requantization scales reordered, and the consumer's weight column for tap s·N + p comes from channel π(p). As ReLU and requantization act per channel and the consumer adds the same INT32 products, the integer output is identical for every input. We permute the outputs of conv1 to conv(L−2) of an L-layer network; the last layer, which can never pay, keeps its input. In the maxout and averaging networks, which keep the maximum or mean of maps j and j + N of 2N computed maps, both maps move with π. For N = 45, tap s·45 + p is in bank (s + p) mod 4: the Original layout already has the skew's banks.

Fitting. Relabeled windows are original windows with permuted columns, so each layer is fitted alone. The recipe that produced Frozen draws 2048 windows per fitted layer from Set12 01.png (Cal-1; σ = 25, one noise draw), minimizes their mean of (1) at W = 4, and searches from the Original with 400 proposals, each swapping two channels of different classes (position mod 4) and kept only if the objective falls (7.1–28.8 s per network on a 4-core x86 server). A second layout of the board network, Multi-start, adds Set12 02–03, two noise draws per image (Cal-3) and four random starts (2005 objective evaluations per layer against 401); it was chosen for the board after its BSD68 counts had been seen.

IV. Experimental Setup

Networks and images. Ten trained integer DnCNN-style networks [9] of six architectures, for Gaussian noise of σ = 25 on 7-bit inputs: A, 7 layers of 32 channels; B, 10 layers of 64 channels behind a 2×2 space-to-depth input; B-32, the same with 32 channels, with ReLU (training seeds s1–s3), maxout over pairs of 64 computed outputs (s1–s3) or their mean (s2); and a 45-channel ReLU variant. All but A had quantization-aware training. The board network, B-32 ReLU s1, reaches 29.54 dB on Set12 05.png; 37–55% of its hidden layers' window values are nonzero. Test images: Set12 [9] and BSD68 [10], 80 images, one noise draw each; savings exclude the calibration images unless stated.

Board. The W = 4 bitstream for the XC7Z020 closes timing at 100 MHz (worst negative slack +0.130 ns), uses 11,151 LUTs (21%), 71 of 140 BRAM36-equivalent tiles and no DSP slice, and ran at the board preset's 50 MHz. One bitstream ran all three layouts; per layer the driver checks the output against the integer model and reads the cycle counter CYCLES and the DMA-wait counter IN_STALL. On 05.png the Original ran once as a

canary, then each layout three times in alternation. These board cycles are transcribed from run notes, each checked by a script against its cited line; the raw reports are not yet committed, except an earlier run of the Original. All other cycle numbers are counts over the windows of the host emulation, which reproduces every network's reference PSNR on all 80 images.

What was fixed when (GitHub push times, UTC). The fill-and-drain rule (6dc0e60, 10-17 13:15) preceded both exports but was read off board runs that include the board network's Original, whose layers are thus in sample; Frozen's per-layer count savings on 05.png (9cf9c79, 10-17 18:38) preceded the board runs; a protocol (c2299e2, 10-18 18:32) fixed the recipe, the evaluation, condition (2) per layer and four hypotheses before any fit on the other nine networks. The skew (2928d28) was counted after the results, outside that protocol.

V. Results

A. Board Measurement

On Set12 05.png, Multi-start lowered the board cycles by 7.60% and Frozen by 7.02% (Table I). The repeats gave the same numbers, except 12 more cycles in Frozen's third run, all in conv10, where IN_STALL was also 12; none of the 90 layer runs of the repeats had an output mismatch. In all 63 comparisons (every layer of Original and Multi-start, sums and totals of canary, repeats and the earlier run), CYCLES − IN_STALL equaled the exact prediction; as the repeats are deterministic and conv1 and conv10 see the same windows in every layout, nine are independent and out of sample, all on one image: Multi-start's conv2–conv9 and Frozen's conv2–9 sum. In counts on 05.png, relabeling lowered the fullest bank's mean nonzero count in every relabeled layer (conv9, Multi-start: 43.04 to 35.10), and Multi-start removed 66.2% and Frozen 61.1% of the saving of perfect balance (11,761,449 cycles, −11.49%).

TABLE I. BOARD CYCLES OF B-32 RELU S1 ON SET12 05.PNG AT W = 4 (REPEAT 1, TRANSCRIBED), EACH EQUAL TO ITS EXACT PREDICTION, AND SAVINGS FROM THEM.

Layout	conv2–9	Total	Saving (%), conv2–9/total
Original	11,421,202	13,289,030	—
Frozen	10,487,986	12,355,814	8.17/7.02
Multi-start	10,410,591	12,278,419	8.85/7.60

B. Study of Ten Networks

The protocol fixed two checks and two sign predictions. H0: our code reproduced the earlier Frozen layout in all eight layers, and folding it reproduced the Frozen export byte for byte. H1: wherever (2) holds, at W = 1 on all networks and at W = 2 on A and B-32 ReLU s1–s3, the relabeled count equaled the original on all 80 images. H2: the pooled W = 4 saving over the 79 images was positive on all ten networks, 1.09–10.80% (Fig. 2; the sign was known for B, maxout s1 and the board network). H3: the pooled W = 2 saving was positive on all six networks where (2) allows it, 3.36–7.37% on five and 0.022% on the 45-channel network, where 16 of 632 image-layers (8 fitted layers × 79 images) got worse (at most 3 elsewhere). No image counted more cycles on any network at any W.

In emulation every folded network reproduced the original's PNG bytes on Set12 05.png and BSD68 test001.png. Maxout s1 gained only 3.50% at W = 4; without per-network headroom, a weak fit cannot be told from a small opportunity. As the protocol's search routine rejects N = 45, that network was fitted with a general bank rule (identical layouts on the 69 layers where both apply).

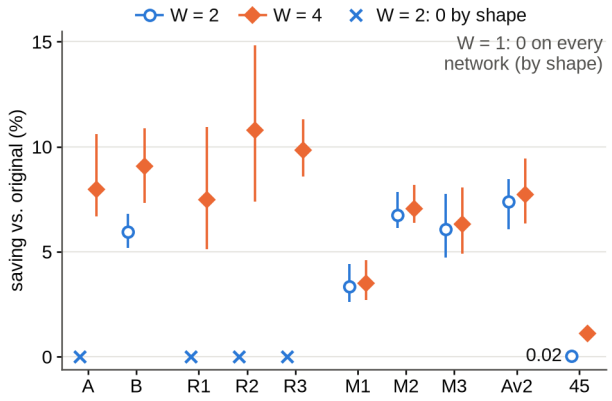


Fig. 2. Pooled saving of the recipe over the 79 images other than the calibration image (count, conv1–convL); lines: per-image range; ×: (2) holds in every layer, saving exactly 0. A: 7 × 32; B: 10 × 64; R1–R3, M1–M3: B-32 ReLU and maxout, seeds s1–s3 (R1: board network); Av2: averaging, s2; 45: 10 × 45 ReLU.

C. Eighty Images and Baselines

At W = 4 (Fig. 3), Frozen saved 5.13–10.94% per image on the 79 images other than its calibration image (pooled 7.46%), Multi-start 5.62–11.19% on the 77 other than its three (pooled 8.11%); perfect balance would save 7.99–16.62% (pooled 12.02%). From the smallest saving, 05.png ranks 34th (Frozen) and 32nd (Multi-start) of 80.

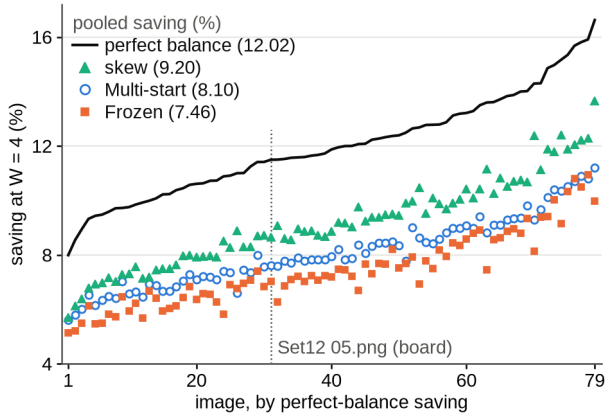


Fig. 3. Per-image W = 4 saving of the board network (count, conv1–10) on the 79 images other than Set12 01.png, sorted by the perfect-balance bound; legend: pooled saving over these 79, which include Multi-start's calibration images Set12 02–03 (not marked). Counts, not board runs.

Twenty random layouts with eight channels per class saved −5.43% to +1.73% pooled, so savings against the Original understate those against a random order. Six layouts dealing channels by their mean nonzero count on the calibration windows, after SparTen's density sort [4], captured 21–33% of the perfect-balance saving (Table II: the best on Cal-1), a Sense-like sort by each channel's nonzero count in the test image at most 40.7%, the deployable fitted layouts 62–69%; this compares procedures and does not isolate co-activation from density. Multi-start's 0.63-point lead over Frozen on the 77 images (8.11% against 7.48%) comes from Frozen's seed (the worst of five), five times the budget and Multi-start's data, not its restarts: with equal data and budget, plain greedy search (Table II) beat Multi-start on four of five seeds.

Skew (counted after the study, not run on the board). On the inputs of conv2–conv9 the skew saved 9.20% pooled over the 79 images (0 at W = 1 and 2, as (2) requires) against 7.46% and 8.10% for the relabelings, and fewer cycles than both on every image (Fig. 3; on the 77 without Multi-start's calibration images, 9.22% against 8.11%); on 05.png it saved 8.64%, and in the integer oracle (4096 windows per layer of 05.png) it left every output unchanged. In counts on this network, the relabeling is thus not the best static layout. Whether a relabeling fitted on top of the skew adds to it is open and will be tested under a new pre-registration; uncommitted exploratory counts of this combination, and of the skew on B and maxout s1, were seen and are not used.

TABLE II. LAYOUTS OF B-32 RELU S1 AT W = 4 (COUNT, 79 IMAGES; † 77). P.I.: USES THE TEST IMAGE. RANDOM: THE BEST OF 20 ON THESE IMAGES. SHARE: (ORIGINAL − LAYOUT)/(ORIGINAL − PERFECT BALANCE), CYCLES SUMMED OVER THE IMAGES.

Layout	Pooled (%)	Share (%)
Random, best of 20	1.73	14.4
Density balance (after [4]), Cal-1	3.69	30.7
Sense-like sort [3] (p.i.)	4.89	40.7
Frozen (board), Cal-1	7.46	62.1
Multi-start (board), Cal-3†	8.11	67.4
Greedy, 5-seed mean, Cal-3†	8.24	68.5
Recipe fitted per image (p.i.)	8.45	70.3
Skew, data-free (post hoc)	9.20	76.6
Perfect balance (bound)	12.02	100

VI. LIMITATIONS

• **Board evidence.** The relabelings were measured on one network and one image, and the fill-and-drain constants come from board runs of four networks on the same image. This bitstream's IN_STALL misses DMA starvation at window boundaries (fixed in later RTL), so the exact match rules out only hidden stalls that differ between layouts; that the rule also holds in RTL simulation, which has no DMA, suggests the constants contain none.

• **Scope.** Only core cycles are reported; host time, energy and FIFO-decoupled banks were not measured. The skew has no export or ten-network count yet.

• **Statistics.** Only $\sigma = 25$ with one noise draw per image and no confidence intervals; BSD68 was part of the networks' development.

VII. CONCLUSION

On a zero-skipping core with lockstep banks, the layer shapes tell where the tap-to-bank layout can matter, and a folded relabeling changed the cycles measured on one image exactly as predicted, with identical output. It lowered the pooled count on all ten networks, but on the measured denoiser a data-free skew, counted and not yet run, saved more. Scripts, layouts, per-image counts and the source of every number are in bankperm/ of https://github.com/qewr1234/FPGA.

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